Scalability & Future Plan

CrimeVision | Intelligent Multimodal Forensic Analysis Platform

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Project Name	CrimeVision	**Document status**	Project-aligned draft

Purpose. This document describes the current prototype's constraints and a realistic path to scale it. The current implementation should be treated as a single Flask application with filesystem-based storage and CPU-based model inference, not as a production-scale deployment.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Single application process; no background job queue is confirmed.	Long video/audio analysis can block requests or slow other users.	High
2	Filesystem-based evidence/output storage; no database-backed case store is confirmed.	Harder to manage concurrent access, retention, backups, and growth across hosts.	High
3	Authentication, role-based access, audit trail, and production hardening are not confirmed.	Sensitive evidence could be exposed if the development server is shared or publicly exposed.	High

Scalability Plan

Aspect	Current state	Proposed upgrade / solution
User load	Single Flask app; concurrent workload limits have not been benchmarked.	Use a production WSGI server behind a reverse proxy; add a task queue and worker processes for long-running media analysis.
Data storage	Local folders for evidence, processed data, outputs and reports.	Use a database for case/evidence metadata and managed object storage for files; implement backups, retention and integrity checks.
Performance	CPU-only inference in the verified development environment; no benchmark results available.	Profile representative workloads, cap file size/duration, sample video frames where appropriate, cache reusable results, and evaluate GPU workers if justified.
Security	Development prototype; production access controls are not confirmed.	Add authentication, authorization, CSRF protections where applicable, upload validation, safe filenames, audit logs, HTTPS and secret management.

Future Roadmap

Phase	Planned enhancement	Target timeline	Expected impact
Phase 1	Stabilize core workflows; add automated tests and document setup.	Near term	More repeatable local demos and easier setup.
Phase 2	Persistent case metadata, evidence hashes, and audit events.	After core workflows	Better traceability and data management.
Phase 3	Background processing, job status, and controlled concurrency.	Mid term	Improved responsiveness for larger files and multiple users.
Phase 4	Authentication, hardened deployment, monitoring, and measured performance evaluation.	Before shared/operational deployment	Safer hosting and evidence-based capacity planning.

Note: timelines are proposed phases, not committed delivery dates. No throughput, accuracy, or latency figures are claimed because formal performance measurements have not been supplied.